



Morning Brief

Ludovici · Tuesday, June 16, 2026

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By Stephanie Ludovici

A personal learning and awareness brief: AI governance, cybersecurity, the federal front, the AI industry landscape, project status, five new project ideas, and four training items. Built to keep me current and sharp.

AI governance & policy

- **The EU AI Office's draft Article 50 transparency *guidelines* sharpen what providers and deployers must actually do — and they bite harder than the voluntary Code.** Separate from the Code of Practice on marking AI-generated content (covered yesterday), the Commission's draft *guidelines* detail how the four Article 50 obligations apply to interactive systems, providers of synthetic-content generators, emotion-recognition/biometric-categorization deployers, and deepfake/AI-text deployers. The notable line: **deepfake labeling applies even with no intent to deceive** — content that looks or sounds like a real person must be labeled regardless. The stakeholder consultation closed June 3. *Why it matters:* this is the binding-interpretation layer beneath the marking rules, and it maps directly onto the AIM framework's v2.1 **AI-output provenance marking (§7.1)** and **output classification (§5.2.1)** — a concrete benchmark for machine-

readable marking vs. human-facing labeling. [Greenberg Traurig · Bird & Bird](#)

- **OpenAI publishes a "blueprint for democratic governance of frontier AI" — and pointedly diverges from the White House (Jun 2–3).** The policy paper argues the U.S. should build a durable *federal* framework anchored on **CAISI** as the government's primary frontier-AI-safety institution, leaning on the emerging consensus in state frontier-safety laws rather than the lighter-touch, voluntary posture of the June 2 EO. *Why it matters:* a frontier lab is now proposing the specific institutional plumbing — who evaluates, who sets standards — that an IC-facing framework would also depend on, and the CAISI-centric design is the same evaluation-governance backbone I track. [OpenAI](#) · [SiliconANGLE](#)
- **The June 2 EO's cybersecurity machinery starts to take shape — Treasury's AI cybersecurity clearinghouse moves from text to build.** *Promoting Advanced AI Innovation and Security* directs Treasury to stand up an **AI cybersecurity clearinghouse within 30 days** (after consulting the NCD, DHS, DoD, NSA, and CISA) to coordinate vulnerability scanning, validation, and patch prioritization; the EO also empowers CISA to issue Binding Operational Directives. Treasury Secretary Bessent publicly defended the order, stressing the cooperation is **recommended, not mandated** and that the EO creates no licensing or pre-clearance requirement. *Why it matters:* this is the federal "voluntary-but-real" model of AI-cyber governance forming in real time — the same recommend-vs-mandate tension the AIM framework has to resolve for IC controls. [FedScoop](#) · [Federal News Network](#)

Cybersecurity & threats

- **CISA adds two KEVs (Jun 8, 2026), led by an actively-exploited flaw in an AI gateway.** The additions are a **BerriAI LiteLLM command-injection** flaw (CVE-2026-42271) and a **Check Point Security Gateway** improper-authentication flaw (CVE-2026-50751). LiteLLM is a widely deployed LLM proxy/gateway — exploited command injection in the AI control plane is squarely on the AIM threat model (the infrastructure *around* models is now the attack surface, not just the models). Patch the

gateway and the perimeter VPN/gateway together. [CISA KEV \(Jun 8\)](#)

- **CISA adds two KEVs (Jun 15, 2026) — a *second* Catalyst SD-WAN Manager flaw in a week.** A Cisco Catalyst SD-WAN Manager directory/path-traversal flaw (CVE-2026-20262) and a LiteSpeed cPanel plugin symlink-following flaw (CVE-2026-54420) are now flagged as exploited. This is the second Catalyst SD-WAN Manager KEV in seven days (after Jun 9's CVE-2026-20245) — the network management plane of federal/enterprise fabric is under sustained, repeated pressure and belongs at the top of the patch queue. [CISA KEV \(Jun 15\)](#)
- **CISA adds one KEV (Jun 12, 2026): a missing-authentication flaw in Oracle PeopleSoft.** Oracle PeopleSoft Enterprise PeopleTools carries a missing-authentication-for-critical-function bug (CVE-2026-35273). PeopleSoft underpins HR/finance/ERP across many federal and large enterprises, and "missing auth for a critical function" means pre-authentication exposure — confirm no internet-facing PeopleSoft instances sit inside the boundary unpatched. [CISA KEV \(Jun 12\)](#)

On the Federal Front

- **Commerce/BIS orders Anthropic to suspend its top models on national-security grounds (Jun 12).** The Bureau of Industry and Security, in a directive signed by Commerce Secretary Lutnick, ordered Anthropic to suspend all access to **Fable 5 and Mythos 5** by any foreign national worldwide — citing a discovered jailbreak/foreign-access concern. Because nationality-based gating is impractical, Anthropic disabled both models globally. *Why it matters:* this is the first time a federal export-control lever has been used to force a *recall* of a deployed commercial frontier model — a concrete, precedent-setting federal control point on frontier-model deployment, and the live version of the pre-release/deployment-review debate the AIM v3.0 agentic and evaluation overlays anticipate. [Defense One](#) · [Congress.gov CRS IF13217](#)
- **Congress signals it may revisit how DoD governs use of AI after the Anthropic–Pentagon fallout.** Reporting indicates lawmakers are weighing NDAA language on military AI use in the wake of the dispute, which already includes the Pentagon's March designation of Anthropic as a **"supply-chain risk"** (now

under challenge in federal court). *Why it matters:* the FY26/FY27 NDAA is the vehicle that codifies DoD AI-evaluation and governance mandates, and a vendor-specific blowup is exactly the kind of event that reshapes statutory text — the authoritative federal counterpart to the framework work I track. [Nextgov/FCW](#)

- **DoD leans harder on AI for contested logistics (early June).** Senior defense officials are pushing AI and related tech to address contested-logistics risk as military supply chains face growing threat of adversary disruption. *Why it matters:* logistics is a decision-support, high-stakes AI use case — exactly the Tier-1/Tier-2 territory where evaluation, oversight, and provenance controls matter most, and a useful real-world test of risk-proportionate governance. [DefenseScoop](#)

AI Industry Landscape

- **Anthropic disables Fable 5 and Mythos 5 worldwide and disputes the order (Jun 12–14).** Forced to comply with the Commerce directive, Anthropic argues that recalling a model "deployed to hundreds of millions of people" over a "narrow potential jailbreak" sets a standard that "would essentially halt all new model deployments for all frontier model providers." *Why it matters:* the competitive cost of being on the wrong side of a national-security finding is now concrete and global — frontier vendors must weigh deployment speed against the risk of a sudden federal recall. [Time](#) · [Fortune](#)
- **The competitive ripple: Anthropic's federal exclusion deepens while rivals hold the classified cohort.** With the March supply-chain-risk designation now compounded by the model suspension, Anthropic sits outside the eight-firm DoD classified-AI cohort (OpenAI, Google, Microsoft, NVIDIA, AWS, Oracle, SpaceX, Reflection per the May 1 announcement) at exactly the moment SASC's proposed cyber/AI reorganization could redraw where each vendor plugs into DoD. *Why it matters:* federal access and the commercial frontier race are diverging, and the org chart is becoming part of the competitive map. [SiliconANGLE](#) · [DefenseScoop](#)
- **OpenAI turns governance into positioning.** Releasing a federal-framework blueprint that elevates CAISI — just as a chief rival absorbs a federal blow — signals that frontier labs are now competing on *governance posture*, not only capability. *Why*

it matters: whoever's preferred institutional design becomes the federal default gains a durable structural advantage; this is lobbying-by-whitepaper for the rules an IC framework would inherit. [OpenAI](#) · [SiliconANGLE](#)

Project status

- **NetMon-Server** (*most active* — `netmon-server.ps1` and `CHANGELOG` edited Jun 14–15) — Single-file Windows PowerShell SIEM/network monitor, now at **v0.41.0**, mid-way through the "SIEM re-layout" (Phase A): the flat 6-tab nav has been restructured into a grouped **21-screen** navigation across five clusters (Monitor / Triage / Investigate / Detect & Harden / System), with Phase-B/C screens shipped as honestly-labeled stubs. Loopback-only, built-in cmdlets only, advisory-only philosophy preserved. *Next action*: complete the Phase-A content pass on the stubbed re-surface screens (Dossier, Copilot, Rules, Report, Self-Health) before starting Phase-B (Net Map, Incidents, Log Search). Today's KEV cluster on management-plane gear (Cisco SD-WAN Manager) is a good prompt to make sure NetMon's exposure/advisory views flag management-interface reachability.
 - **MongoDB (Personnel Vetting research)** (*tracker updated Jun 12*) — Working draft of a research paper mapping the personnel-vetting data/system flow (DISS, NBIS eApp, PIPS, Mirador, CATS, Scattered Castles) as two connected cycles (initial vetting + continuous vetting) rather than a linear pipe. System inventory and process flow are agreed. *Next action*: finalize the DFD figures (`fig2_two_cycles.svg`, `fig4_dual_path.svg`) and lock the framing points, then draft the narrative section.
 - **AI Framework** (*roadmap reviewed*) — v2.1 framework is shipped; `V3.0-ROADMAP-FOLLOWUP.md` tracks the deferred agentic-AI and GenAI-groundedness overlays, both flagged as needing an evidence verification pass before any control is drafted. *Next action*: given today's federal export-control precedent against a frontier model, capture a one-line note in the v3.0 tracker that the "deployment control / pre-release review" theme now has a concrete federal data point worth citing when the agentic overlay's human-confirmation and blast-radius controls are specified.
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Five project briefs

1. Deep Koopman autoencoder for data-driven forecasting of nonlinear dynamics

- **Domain:** Fluid dynamics / nonlinear dynamical systems ·
Language(s): Python (PyTorch)
- **Problem statement:** Many scientific systems (wakes, convection, reacting flows) are nonlinear, but linear models are far easier to forecast, control, and reason about. Koopman theory says a nonlinear system becomes *linear* in a lifted space of observables — yet choosing those observables by hand is the open problem. Classical DMD approximates the Koopman operator with linear measurements and breaks on strongly nonlinear transients. The gap is a learned, compact observable space that stays predictive over long horizons.
- **Objective(s):** Train an autoencoder whose latent coordinates evolve under a single learned linear operator K , giving accurate multi-step forecasts and interpretable Koopman modes/eigenvalues for a canonical flow.
- **Questions to answer:**
 - Does a learned observable space beat plain DMD on multi-step forecast error for a nonlinear flow?
 - How many latent dimensions are needed before forecast skill saturates?
 - Do the recovered Koopman eigenvalues match the system's known oscillation/decay rates?
- **Proposed method/approach:** Build snapshots from a 2-D cylinder wake (or Kuramoto–Sivashinsky); encoder ϕ maps state $\rightarrow$ latent z ; enforce linear latent dynamics $z_{t+1}=K z_t$ via a small linear layer; decoder reconstructs state. Train with reconstruction + one-step + multi-step prediction losses; diagonalize K for modes; compare against exact DMD and an LSTM baseline on held-out time.
- **Tools:** PyTorch, NumPy, SciPy; pydmd for the DMD baseline; Johns Hopkins Turbulence Database or self-simulated K–S (py-pde). One consumer GPU (or CPU for K–S).
- **Scope:** *In scope* — one 2-D system, forecasting + mode analysis, DMD/LSTM baselines. *Out of scope* — 3-D turbulence, control synthesis, experimental PIV. *Deliverables* — training

notebook, eigenvalue/mode plots, forecast-error-vs-horizon curves, write-up.

- **Estimated duration to complete:** 3–4 weeks part-time.
- **Sources:** [Lusch, Kutz & Brunton, *Nature Communications* 2018](#)
· [PyDMD docs](#)

2. Double machine learning for heat–health causal attribution

- **Domain:** Environmental epidemiology / causal inference ·
- **Language(s):** R (DoubleML, grf), Python EconML cross-check
- **Problem statement:** Heat–mortality associations are confounded by season, humidity, air quality, and demographics. Naïve regressions mis-state the effect; plain ML predicts well but biases effect estimates. Double/debiased ML uses flexible learners for nuisance functions while preserving valid inference on the causal effect — the right tool for attribution that must survive scrutiny.
- **Objective(s):** Produce a debiased estimate (with honest CIs) of the effect of heatwave days on excess mortality, plus heterogeneity by region/age via causal forests.
- **Questions to answer:**
 - What is the average treatment effect of a heatwave day on daily mortality after debiasing?
 - How much does the estimate move between a naïve GLM and DML?
 - Which subpopulations show the largest heterogeneous effects?
- **Proposed method/approach:** Assemble daily mortality with temperature/humidity/PM2.5 covariates; define treatment = heatwave indicator; fit nuisance models with gradient boosting; apply partially-linear DML with cross-fitting; estimate heterogeneity with generalized random forests; validate with placebo/negative-control outcomes and unmeasured-confounding sensitivity analysis.
- **Tools:** R DoubleML, grf, ranger/xgboost; Python EconML cross-check; CDC WONDER mortality
- NOAA/ERA5 temperature + EPA AQS PM2.5. Laptop-class compute.
- **Scope:** *In scope* — one region, ATE + heterogeneity, robustness checks. *Out of scope* — individual-level clinical data, full spatiotemporal causal modeling. *Deliverables* —

reproducible R pipeline, effect estimates with CIs, heterogeneity maps, sensitivity report.

- **Estimated duration to complete:** 3–4 weeks part-time.
- **Sources:** [Chernozhukov et al., *Econometrics Journal* 2018](#) · [DoubleML for R](#)

3. Geometric deep learning on cortical surfaces for brain-age prediction with uncertainty

- **Domain:** Neuroscience / neuroimaging · **Language(s):** Python (PyTorch Geometric)
- **Problem statement:** The cortex is a folded 2-D sheet; flattening it to a grid for a standard CNN discards geometry. "Brain age" (predicted minus chronological age) is a sensitive biomarker, but reliable use needs both surface-aware models and calibrated uncertainty — most pipelines lack both.
- **Objective(s):** Train a mesh/graph neural network on cortical-surface features (thickness, curvature) to predict age, with calibrated predictive intervals so the brain-age gap carries error bars.
- **Questions to answer:**
 - Does a surface-aware GNN beat a tabular/ROI-summary baseline on age MAE?
 - Are the predictive intervals calibrated on held-out subjects?
 - Which cortical regions drive predictions (saliency)?
- **Proposed method/approach:** Use FreeSurfer surfaces on a standard mesh template; build per-vertex feature graphs; train a GraphSAGE/Cheb-conv regressor; add uncertainty via deep ensembles or MC-dropout; post-hoc calibrate intervals with split conformal; compare against ridge/XGBoost on ROI summaries; compute vertex saliency.
- **Tools:** PyTorch Geometric, nibabel, nilearn; IXL or an OpenNeuro release with FreeSurfer derivatives. One consumer GPU (8–12 GB).
- **Scope:** *In scope* — healthy-aging cohort, age regression + UQ + saliency. *Out of scope* — clinical diagnosis, longitudinal modeling, large-scale raw preprocessing. *Deliverables* — model code, calibration plots, brain-age-gap distributions, region-importance maps.
- **Estimated duration to complete:** 4–6 weeks part-time.

- **Sources:** [Bronstein et al., "Geometric Deep Learning," arXiv: 2104.13478](#) · [Cole & Franke, *Trends in Neurosciences* 2017](#)

4. Bayesian neural ODE for continuous-glucose physiology with quantified process noise

- **Domain:** Digital health / physiology · **Language(s):** Julia (SciML: DiffEqFlux, Turing), Python cross-check
- **Problem statement:** CGM traces are irregularly sampled, noisy, and governed by partly-unknown physiology (meals, insulin, circadian effects). Black-box sequence models ignore mechanistic structure; pure compartment models miss real dynamics. A neural ODE augmenting a minimal physiological model, fit in a Bayesian way, can interpolate, short-horizon forecast, and report honest uncertainty.
- **Objective(s):** Fit a hybrid (mechanistic + neural) ODE to CGM data with Bayesian inference, yielding forecasts with credible intervals and a quantified split between process and observation noise.
- **Questions to answer:**
 - Does augmenting a minimal glucose–insulin ODE with a neural term improve short-horizon forecasts?
 - How are process noise and measurement noise apportioned by the posterior?
 - Do credible intervals achieve nominal coverage on held-out windows?
- **Proposed method/approach:** Start from a minimal Bergman-style glucose ODE; add a small neural term for unmodeled dynamics (the focus is Bayesian UQ, not system ID); define a state-space likelihood; infer parameters + weights with NUTS (Turing.jl) or SGLD; evaluate interpolation, k-step forecast, and interval coverage; compare against a Gaussian-process baseline.
- **Tools:** Julia `DifferentialEquations.jl`, `DiffEqFlux.jl`, `Turing.jl`; Python `torchsde` cross-check; OhioT1DM or an open Tidepool/Nightscout export. Laptop/CPU feasible; small GPU optional.
- **Scope:** *In scope* — single-subject hybrid model, UQ, coverage evaluation. *Out of scope* — closed-loop dosing, multi-subject hierarchical pooling, any clinical claim. *Deliverables* — Julia notebook, posterior summaries, forecast/coverage plots, noise-decomposition table.

- **Estimated duration to complete:** 4–6 weeks part-time.
- **Sources:** [Rackauckas et al., "Universal Differential Equations," arXiv:2001.04385](#) · [SciML / DiffEqFlux docs](#)

5. Closed-loop Bayesian optimization for materials discovery

- **Domain:** Materials science · **Language(s):** Python (BoTorch / Ax)
- **Problem statement:** Brute-force search over composition/process space for a target property is wasteful when each evaluation — even a simulation — is costly. Bayesian optimization closes the loop: a surrogate proposes the most informative next candidate, balancing exploration and exploitation. Doing it on a real materials table teaches the parts that bite (mixed/discrete variables, batch proposals, noisy objectives), not just toy functions.
- **Objective(s):** Build a closed-loop BO agent that reaches a target material property in far fewer evaluations than random/grid search, with honest regret curves.
- **Questions to answer:**
 - How many evaluations does BO need vs random search to hit the target property?
 - Does batch (q-EI) BO beat sequential under a fixed budget?
 - How sensitive is performance to the acquisition function and surrogate kernel?
- **Proposed method/approach:** Treat a public materials dataset as the oracle; featurize compositions (matminer); fit a GP surrogate; optimize Expected Improvement / UCB and batch q-EI; run repeated trials and report mean regret with confidence bands vs random and greedy baselines; ablate kernel/acquisition.
- **Tools:** Python BoTorch + Ax, GPyTorch, matminer; Matbench or the Materials Project API. Laptop-class compute (small GPU optional).
- **Scope:** *In scope* — single target property, sequential + batch BO, regret benchmarking. *Out of scope* — multi-objective Pareto search, wet-lab/DFT execution, generative design. *Deliverables* — BO loop code, regret-vs-budget plots, acquisition/kernel ablation, summary memo.
- **Estimated duration to complete:** 2–3 weeks part-time.

- **Sources:** [Frazier, "A Tutorial on Bayesian Optimization," arXiv: 1807.02811](#) · [BoTorch docs](#)

Training

1. Python — `itertools` for memory-efficient, composable pipelines
`itertools` builds lazy iterators you can chain like Lego, so you stream data instead of materializing giant lists. This matters whenever inputs are large or infinite: a generator pipeline holds one item at a time, keeping memory flat. Four routinely useful tools: `islice` (slice without indexing), `accumulate` (running totals/maxima), `groupby` (group *consecutive* equal keys), and `pairwise` (adjacent pairs, Python 3.10+).

```
from itertools import islice, accumulate, groupby, pairwise

# A lazy source: squares of all non-negative integers (c
def squares():
    n = 0
    while True:
        yield n * n # one value per request, nothing sto
        n += 1

# Take just the first five without building the whole se
first_five = list(islice(squares(), 5)) # [0, 1, 4, 9, 16]

# Running maximum over a stream of readings.
readings = [3, 1, 4, 1, 5, 9, 2, 6]
running_max = list(accumulate(readings, max)) # [3, 3, 4, 5, 9, 9, 9, 9]

# Successive differences via pairwise.
deltas = [b - a for a, b in pairwise(readings)] # [-2, 3, -3, 4, -7, 7, -4]

# Group CONSECUTIVE equal values (note the pitfall below)
runs = {k: len(list(g)) for k, g in groupby("aaabbbccca")}
```

Why it matters: lazy pipelines let one laptop process datasets larger than RAM and compose cleanly into reusable stages. **Common**

pitfall: `groupby` only groups *adjacent* equal keys — it does not sort first. To group an unsorted iterable by a key, call `sorted(data, key=...)` before `groupby(..., key=...)`, or you will silently get many tiny fragmented groups. Reference: [Python itertools docs](#).

2. R — `slider` for principled sliding-window computations

Rolling means, windowed sums, and "last N days" features are everywhere in time series, and hand-rolled loops are slow and error-prone at the edges. The `slider` package (from the tidyverts authors)

gives purrr-style sliding windows with explicit, predictable alignment and edge handling — far cleaner than `ad-hoc zoo::rollapply` for tidy workflows.

```
library(slider)
library(dplyr)

set.seed(1)
df <- tibble(
  day   = as.Date("2026-06-01") + 0:9,
  value = round(runif(10, 0, 10), 1)
)

df <- df %>%
  mutate(
    # Centered 3-day mean; .before/.after set the window
    roll_mean = slide_dbl(value, mean, .before = 1, .after = 1),
    # Trailing 3-day sum; .complete = TRUE returns NA until window is full
    trail_sum = slide_dbl(value, sum, .before = 2, .complete = TRUE)
  )

# Index-aware windowing (e.g., "last 3 calendar days") using
df <- df %>%
  mutate(cal_mean = slide_index_dbl(value, day, mean, .before = 2, .complete = TRUE))
```

Why it matters: `slider` makes the window definition explicit (`.before`, `.after`, `.complete`, `.step`), so feature engineering is reproducible and the train/serve boundary behaves identically. The `*_index_*` family aligns windows to a real date/time index, which avoids the classic bug of treating irregularly-spaced rows as if they were evenly spaced. Reference: [slider package](#).

3. Data-centric — Change Data Capture (CDC): log-based vs query-based

CDC is the discipline of capturing *what changed* in a source system and propagating only those changes downstream, instead of re-copying whole tables. It is the engine behind low-latency replication, incremental warehouse loads, and event-driven architectures. Two dominant patterns:

- **Query-based (polling):** periodically `SELECT ... WHERE updated_at > :last_watermark`. Simple, no special privileges, works anywhere — but it misses hard deletes, depends on a trustworthy `updated_at`, and adds load to the source.
- **Log-based:** read the database's transaction log (WAL/redo/binlog) via a tool like Debezium. It captures inserts/updates/

deletes with low latency and minimal source load, at the cost of more operational complexity and elevated access.

A minimal query-based watermark step:

```
-- Pull only rows changed since the last successful run,
SELECT *
FROM   orders
WHERE  updated_at > :last_watermark    -- :last_watermark
ORDER BY updated_at;
```

Why it matters: choosing the wrong pattern silently corrupts downstream data. Query-based CDC that ignores deletes leaves "ghost" rows in the warehouse; a watermark with second-level granularity can skip rows committed within the same second. Prefer log-based CDC when you need deletes and low latency; pair either pattern with an idempotent upsert/MERGE so re-runs are safe.

Reference: [Debezium documentation](#).

4. AI method — Dynamic Mode Decomposition (DMD) / the Koopman view

What it is. DMD fits a single linear operator **A** that advances a system one time step, $x_{t+1} \approx A x_t$, from snapshot data alone. Stacking snapshots into X_1 (steps 1...m-1) and X_2 (steps 2...m), it computes a low-rank **A** via the SVD of X_1 , then takes the eigendecomposition. The eigenvectors are **DMD modes** (coherent spatial structures); the eigenvalues give each mode's **frequency and growth/decay rate**. It is the practical, finite-dimensional approximation of the **Koopman operator**, which advances *observables* of a nonlinear system linearly.

Intuition / math. Instead of forming the huge $A = X_2 X_1^+$ directly, DMD projects onto the leading r left singular vectors: $\tilde{A} = U_r^* X_2 V_r \Sigma_r^{-1}$, eigendecomposes $\tilde{A} = W \Lambda W^{-1}$, lifts modes back via $\Phi = X_2 V_r \Sigma_r^{-1} W$, and converts discrete eigenvalues to continuous ones with $\omega = \ln(\lambda) / \Delta t$. The state is then reconstructed/forecast as a sum of modes evolving as $\exp(\omega t)$.

When to use it. Reach for DMD when you have regularly-sampled, high-dimensional spatiotemporal data and want interpretable modes plus short-horizon forecasts without training a network — fluids, climate fields, power-grid signals, neural recordings. Its core limitation is linearity: on strongly nonlinear transients it degrades, and a spurious positive growth rate makes forecasts diverge. That failure motivates *deep Koopman* models (project idea #1) that learn nonlinear observables in which the dynamics become linear. Today's

companion notebook demonstrates DMD end-to-end with M&S and T&E. Reference: Kutz, Brunton, Brunton & Proctor, *Dynamic Mode Decomposition* (SIAM, 2016) — [overview](#).

Companion notebook: notebooks/2026-06-16-dynamic-mode-decomposition.ipynb simulates a two-mode spatiotemporal field, fits exact DMD on a training window, then forecasts a held-out window — checking that the recovered frequencies ($\approx 1.3, 3.6$) and growth rates ($\approx 0.0, -0.10$) match the simulated truth.

Signal & Trajectory

Analysis and inference — not reported fact. Each read traces to an item above.

- **If the Commerce/BIS recall of Anthropic's models (Federal Front) holds as precedent, expect "deployment control" to become a first-class governance primitive** — pre-release review (the June 2 EO) plus post-release recall now bracket a model's whole lifecycle. Watch for the next NDAA cycle to codify a standing recall/suspension authority, and treat this as the concrete federal anchor the AIM v3.0 agentic-overlay's human-confirmation and blast-radius controls have been waiting for.
- **If the EU's Article 50 *guidelines* (Governance) land as drafted — deepfake labeling even without intent to deceive — expect the marking-vs-labeling distinction to harden into two separate technical requirements.** That favors machine-readable provenance at generation time over human-facing labels bolted on later; the AIM §7.1 provenance-marking control should be specified to satisfy the stricter generation-time bar, not just the display-time one.
- **If the AI control plane keeps showing up in the KEV catalog (Cybersecurity — LiteLLM command injection), expect "secure the gateway, not just the model" to become standard guidance.** Watch whether CISA issues AI-infrastructure-specific hardening direction; in the meantime, NetMon's exposure/advisory views should explicitly flag reachable AI-proxy and management interfaces.
- **If Anthropic's exclusion from the DoD classified cohort persists (Industry) while OpenAI presses a CAISI-centric blueprint, expect governance posture to become a**

procurement differentiator. Watch the SASC cyber/AI reorganization: where CDAO lands will signal whose institutional design — and whose models — the department standardizes on.

Ledgers updated this run: `publication-ledger.md` (No. 6), `project-ideas/_ledger.md` (5 new ideas), `training-ledger.md` (4 new topics).

Attached: brief (.pdf) + companion notebook (.ipynb) + project briefs (.docx)

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